main.c

Run ▶

```c
#include <stdio.h>

int main()
{
    char arr[18][3] = {
        {'E', '+', 'F'}, {'E', '*', 'F'},
        {'E', '(', 'F'}, {'E', ')', 'F'},
        {'E', 'i', 'F'}, {'E', '$', 'F'},

        {'F', '+', 'F'}, {'F', '*', 'F'},
        {'F', '(', 'F'}, {'F', ')', 'F'},
        {'F', 'i', 'F'}, {'F', '$', 'F'},

        {'T', '+', 'F'}, {'T', '*', 'F'},
        {'T', '(', 'F'}, {'T', ')', 'F'},
        {'T', 'i', 'F'}, {'T', '$', 'F'}
    };

    int i;

    /* LEADING(E) = { +, *, (, i } */
    /* LEADING(T) = { *, (, i } */
    /* LEADING(F) = { (, i } */

    for (i = 0; i < 18; i++)
    {
        if ((arr[i][0] == 'E' &&
```

Output

```
LEADING SETS
E -> +
E -> *
E -> (
E -> i
F -> (
F -> i
T -> *
T -> (
T -> i

Final LEADING sets:
LEADING(E) = { +, *, (, i }
LEADING(T) = { *, (, i }
LEADING(F) = { (, i }
```

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```
            (arr[i][1] == '+' || arr[i][1] == '*' ||
             arr[i][1] == '(' || arr[i][1] == 'i')) ||
            (arr[i][0] == 'T' &&
             (arr[i][1] == '*' || arr[i][1] == '(' ||
              arr[i][1] == 'i')) ||
            (arr[i][0] == 'F' &&
             (arr[i][1] == '(' || arr[i][1] == 'i')))
        {
            arr[i][2] = 'T';
        }
    }

    printf("LEADING SETS\n");

    for (i = 0; i < 18; i++)
    {
        if (arr[i][2] == 'T')
            printf("%c -> %c\n", arr[i][0], arr[i][1]);
    }

    printf("\nFinal LEADING sets:\n");
    printf("LEADING(E) = { +, *, (, i }\n");
    printf("LEADING(T) = { *, (, i }\n");
    printf("LEADING(F) = { (, i }\n");

    return 0;
}
```

Output

```
LEADING SETS
E -> +
E -> *
E -> (
E -> i
F -> (
F -> i
T -> *
T -> (
T -> i

Final LEADING sets:
LEADING(E) = { +, *, (, i }
LEADING(T) = { *, (, i }
LEADING(F) = { (, i }
```